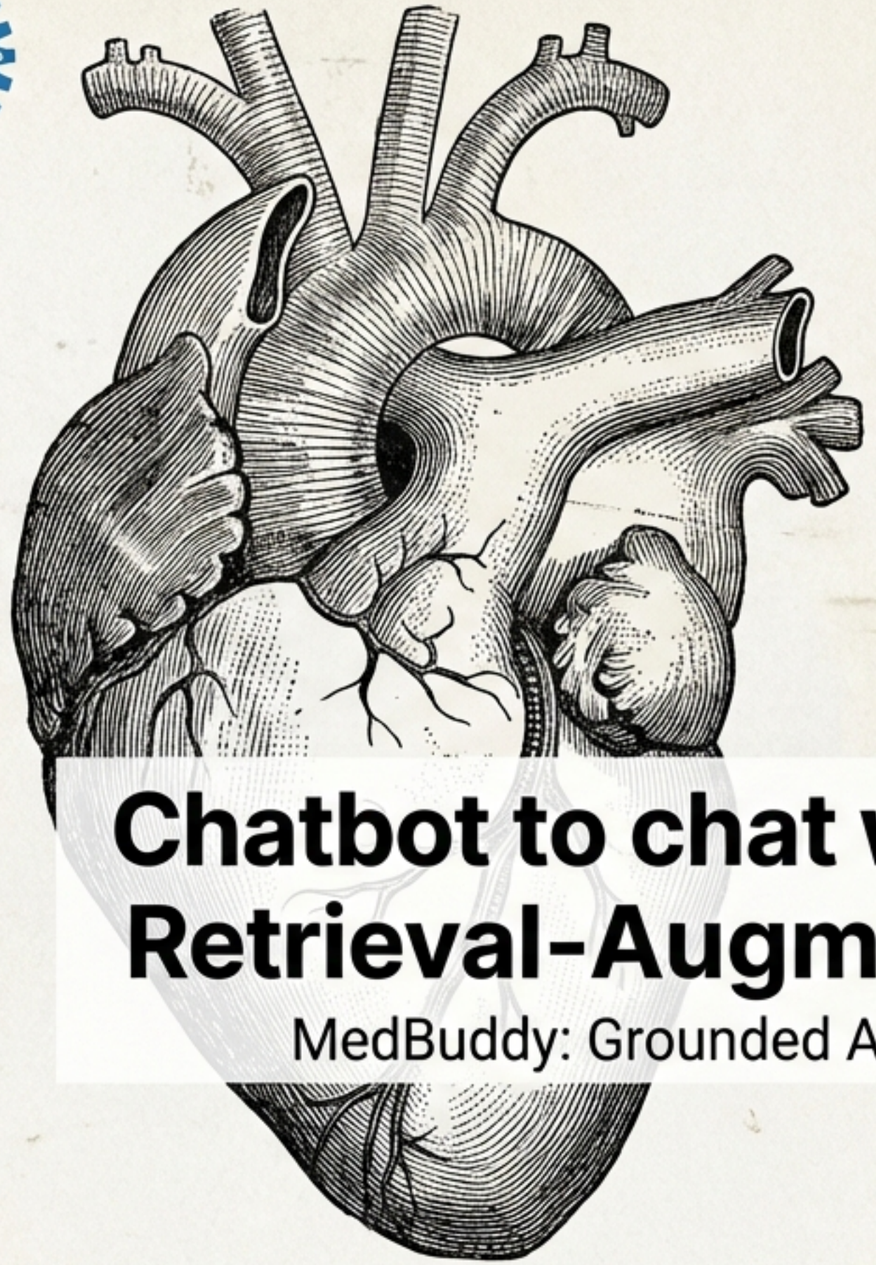
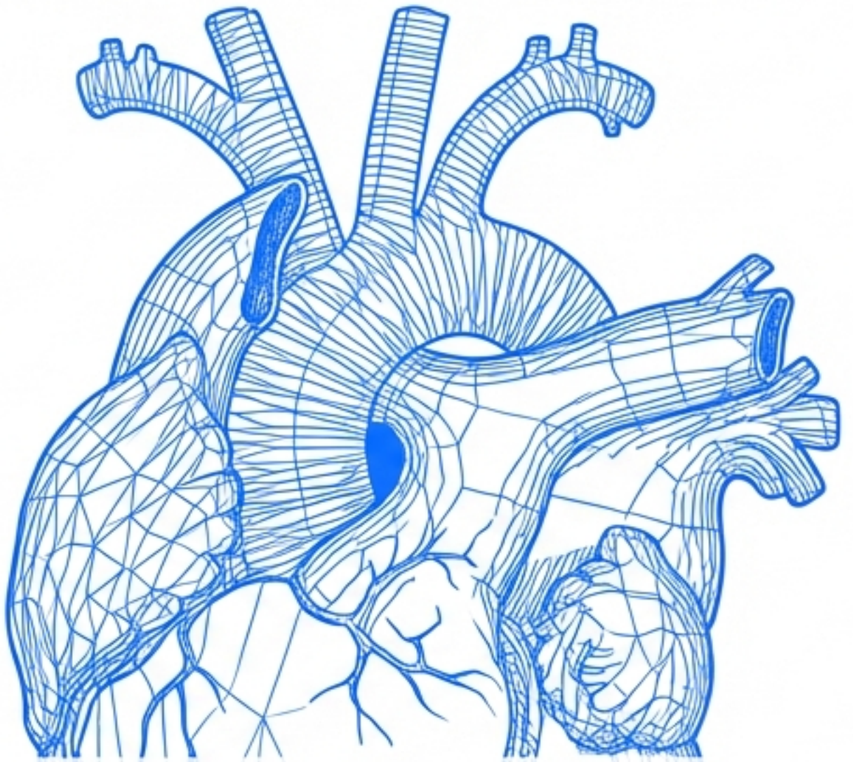




Chatbot to chat with medical books using Retrieval-Augmented Generation Model

MedBuddy: Grounded AI for Evidence-Based Medical Education

Team: Adithya Baiju (RIE23CS006), Akhil Kunduvalappil (RIE23CS011), Devika V S (RIE23CS019), Pavithra P S (RIE23CS037), Prakhar Sharma (RIE23CS039)
Guide: Amrutha S Aravind, Asst. Prof, CSE, RIET
Institution: Rajadhani Institute of Engineering and Technology (RIET)

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Presentation Roadmap

- **Abstract** (The Executive Summary)
- **Introduction** (The Technology Gap)
- **Literature Survey** (Prior Art)
- **Problem Identification** (The Why)
- **Proposed System** (MedBuddy Tech Stack)
- **Architecture** (System Design)
- **Project Plan** (2026 Timeline)
- **Conclusion & Future Scope**
- **References**



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Abstract: Revolutionizing Medical Study

The Mission:

To alleviate the escalating academic workload workload and mental health crisis among medical students by providing instant, accurate access to specialized knowledge.

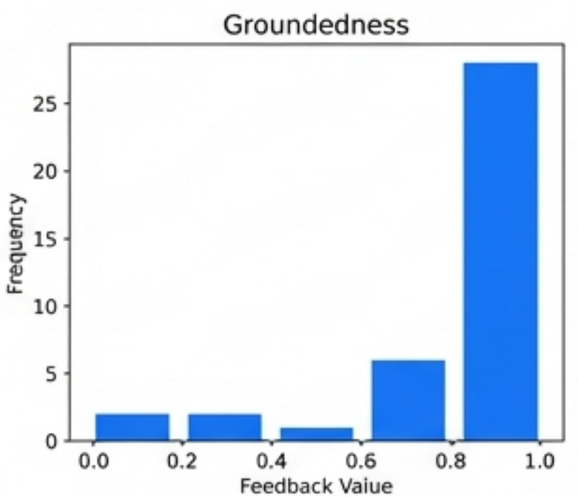
The Technical Solution:

A Grounded AI system using IBM Watsonx APIs and Retrieval-Augmented Generation (RAG) to process datasets like PubMed and Meditrin.

84%
Groundedness Score (Source Reliability)

70%
Reduction in Retrieval Time

65%
Improvement in Learning Efficiency



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The Engine: Large Language Models (LLMs)

Definition

LLMs are advanced AI systems trained on vast volumes of text (medical records, research papers) capable of understanding and generating human-like language.



The Capability

They act as smart assistants that can summarize complex clinical concepts and process natural language queries.

The Limitation

While powerful, generic LLMs (like standard GPT-4) operate on probability, not facts. Without a reference anchor, they are prone to "hallucinations"—confidently inventing medical advice.

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The Solution: Retrieval-Augmented Generation (RAG)

The Mechanism

Standard LLM
(Closed Book)



The AI tries to memorize everything and often guesses (Hallucination).

MedBuddy RAG
(Open Book)



The AI looks up the specific page in the textbook BEFORE answering.

Core Benefit: Eliminates hallucinations. Ensures evidence-based answers that are traceable, reliable, and grounded in medical literature.

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Literature Survey: Existing Solutions

Recent research explores how artificial intelligence chatbots are transforming medical education and public health training. These studies examine both the promising capabilities and critical limitations of AI-powered educational tools.

Paper / Author	Proposed Method	Gap / Disadvantage
'Chatbot to chat with medical books using RAG' - Peri, S. et al. (IEEE, 2024)	Integration of RAG + LLM	Experienced high latency when processing multiple large PDF textbooks.
'Exploring AI Chatbots in Public Health' - Baglivo, F. et al. (2023)	Vaccination training bots	Lacks autonomy; requires heavy professional verification to avoid misinformation.

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Literature Survey: Niche & Cost Barriers

While numerous AI solutions exist, many remain too focused on niche applications or are prohibitively expensive for widespread academic adoption. This survey highlights key examples that underscore the need for a cost-effective, education-centric RAG system like MedBuddy.

Paper / Author	Proposed Method	Gap / Disadvantage
'Ontology Based Personal Assistive App' - Geethalekshmy et al. (2023)	Semantic reasoning for elderly care	Focused on personal logs/daily living, not academic medical education.
'Authenticity in LLMs for Medical Degree Institutions' - Research Trend (2025)	Grounded Generation	Prohibitive infrastructure costs compared to MedBuddy's efficient architecture.

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Problem Identification: The Clinical Knowledge Gap



Academic Stress: Medical students face overwhelming workloads, often spending 70% of study time searching for data rather than synthesizing it.



The Trust Gap: Generic LLMs hallucinate. In medicine, a ‘hallucination’ isn’t a glitch; it’s a potential patient safety risk.



Inefficient Search: Traditional keyword searches (Ctrl+F) fail to understand the context of complex clinical queries (e.g., symptoms vs. diagnosis).



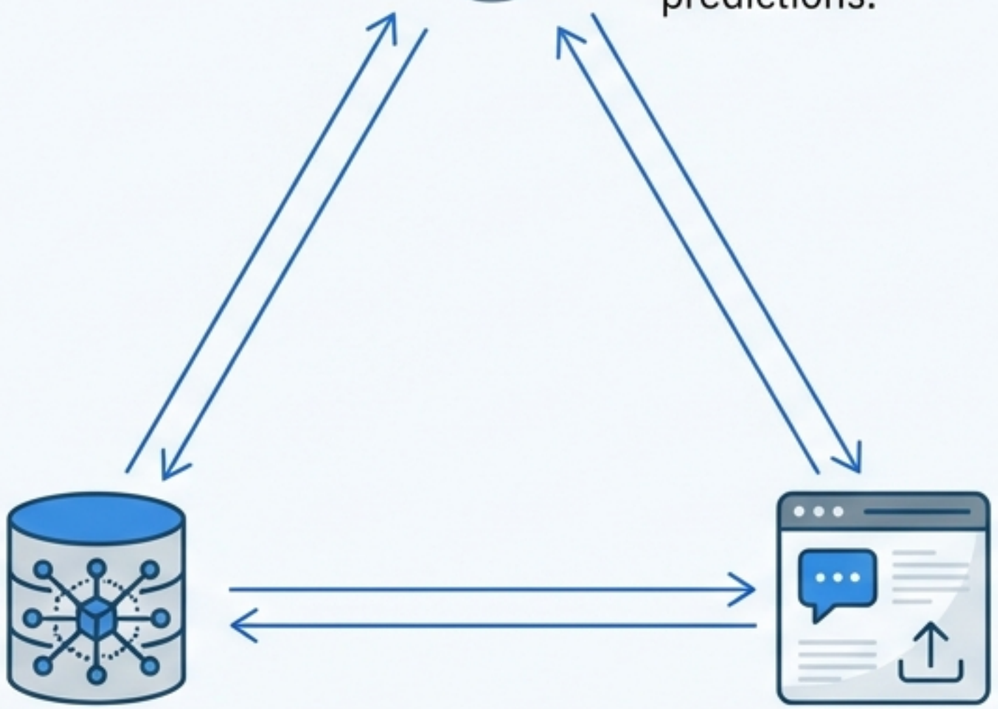
Resource Inequality: High-quality, updated medical textbooks are expensive. MedBuddy democratizes access to this information.

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Proposed System: The MedBuddy Ecosystem

IBM Watsonx.ai Studio

Utilizes AutoAI pipelines to ensure >95% confidence in medical query predictions.



Vector Database

Ingests massive datasets (PubMed, Meditrin), converting text into mathematical embeddings.

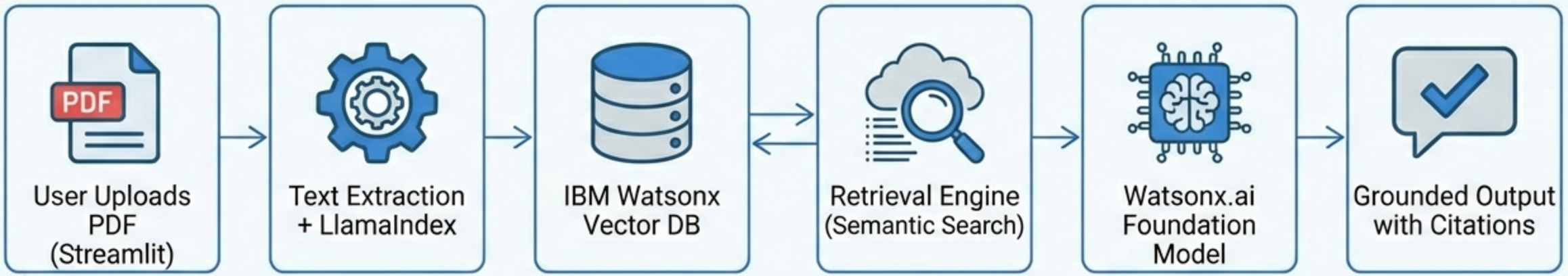
Streamlit UI

Provides a clean, student-friendly dashboard for uploading PDFs and chatting.

Security: HIPAA-Compliant Architecture: Deployed on Watsonx's secure cloud with encrypted data handling.

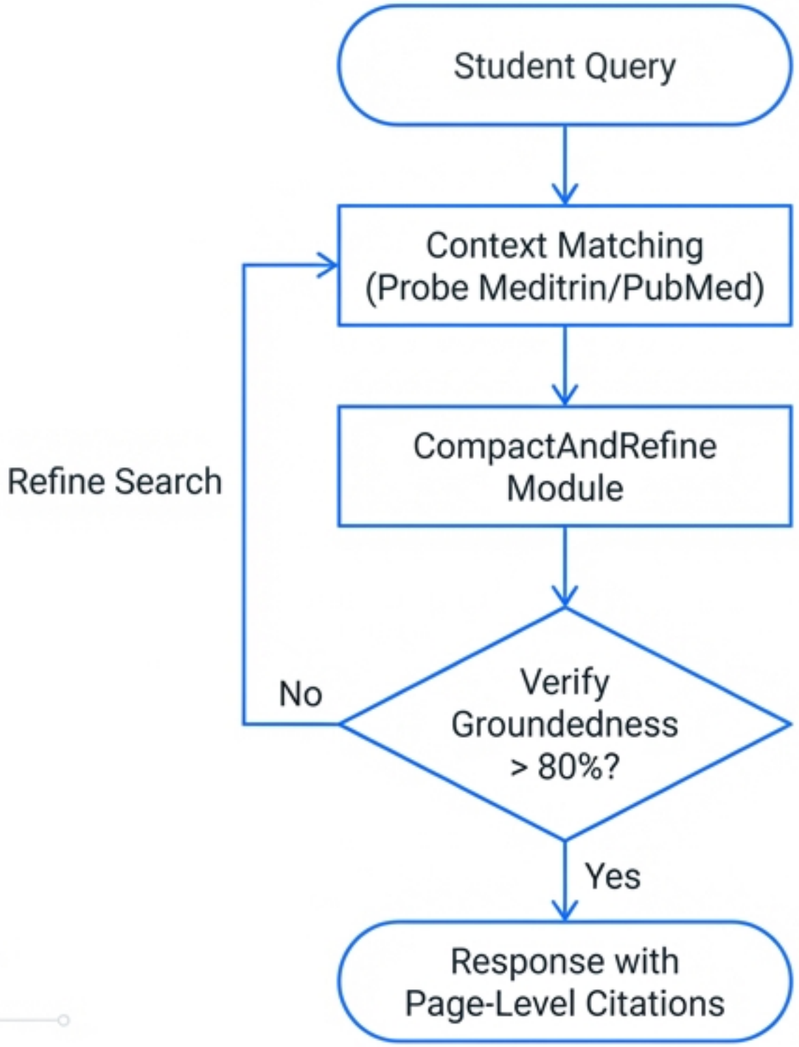
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MedBuddy System Architecture



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Algorithmic Flow: From Query to Citation



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System Requirements

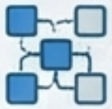
Software Stack



Frontend: React / Streamlit



AI Core: IBM Watsonx.ai Studio

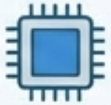


Orchestration: LlamaIndex



Database: PostgreSQL / Vector DB

Hardware Specifications (Client Side)



Processor: Intel i5/i7



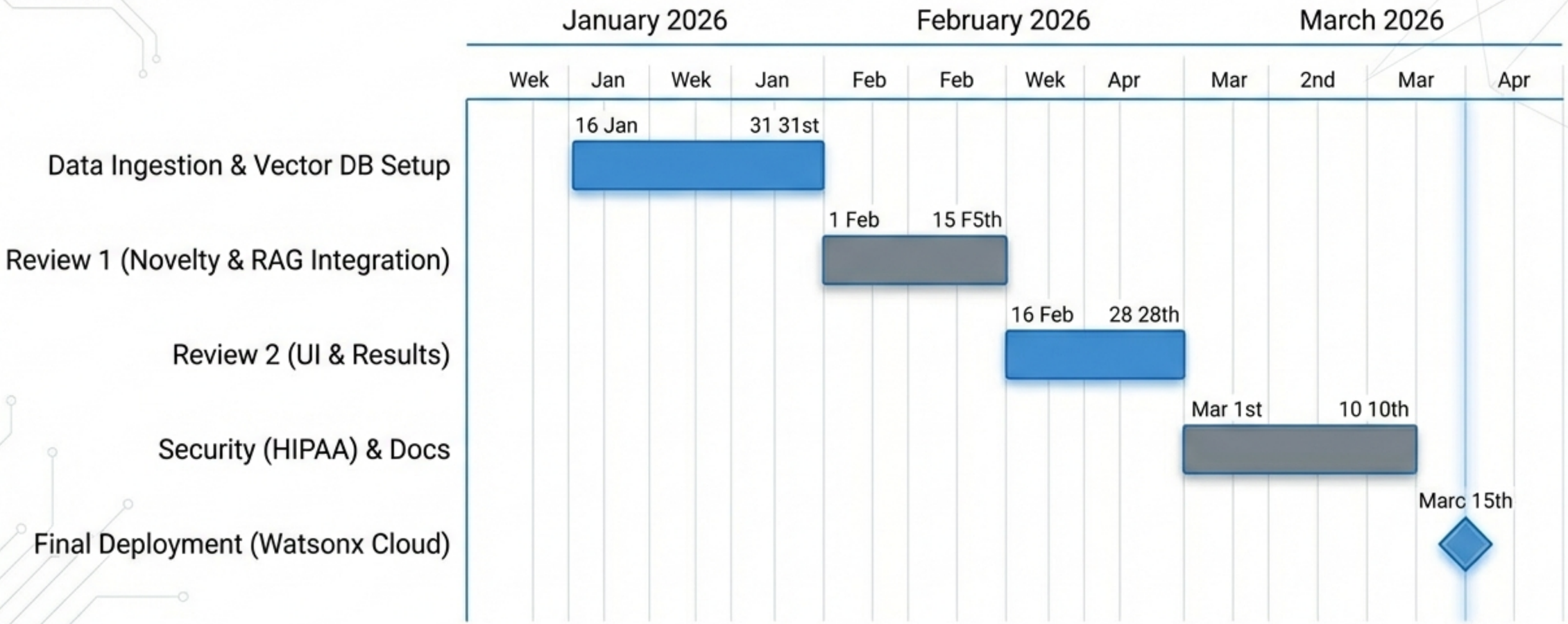
RAM: 8GB Minimum



Storage: 256GB SSD (Minimum)

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Project Plan & Roadmap (2026)



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Conclusion & Future Scope

Immediate Impact:

- 70% reduction in search time.
- Direct mitigation of academic stress.
- Democratization of expensive medical knowledge.

Future Scope: Multimodal RAG



- Multimodal RAG (X-ray/MRI analysis)
- Voice Command Integration
- Clinical Co-Pilot for Doctors

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1. Peri, S. D. B. et al., ‘Chatbot to chat with medical books using Retrieval-Augmented Generation Model,” 2024 IEEE ICINPRO. [Base Paper]
2. Baglivo, F. et al., ‘Exploring the Possible Use of AI Chatbots in Public Health Education,’ JMIR Publications, 2023.
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5. Eysenbach, G., ‘The Role of ChatGPT in Medical Education,’ JMIR, 2023.

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